

Exception	Number of occurrence	Root Cause	Possible Solutions
com.google.cloud.spanner.SpannerException	2	Concurrent modification within a Spanner transaction. Likely due to multiple clients attempting to update the same data simultaneously.	Implement proper transaction isolation levels (e.g., Serializable), use optimistic locking, or redesign the application to avoid concurrent access to critical data. **
java.lang.IllegalStateException	1	Spring application context not initialized. This usually indicates a problem with dependency injection or application startup within the Spring environment.	Ensure Spring is correctly configured and that all dependencies are loaded before attempting to use application context features. Verify the `SpringApplication` is initialized correctly and that any required configuration beans are available. **
Caused by: io.grpc.StatusRuntimeException	1	Transaction was aborted due to concurrency issues within a Spanner transaction. The underlying cause is likely data contention or race conditions during a Spanner operation.	Implement retry logic with exponential backoff for transient errors, consider using a distributed lock to coordinate access to shared resources, or review the transaction design and scope to minimize contention. **
Caused by: java.lang.NullPointerException	2	`customerName` is null, preventing the call to String.length(). This suggests a data validation issue – a customer record is missing a `customerName` field.	Add validation logic to ensure that the `customerName` is not null before proceeding with operations that depend on it. Provide default values or handle null customer names gracefully – perhaps by logging a warning instead of failing the operation. **

Caused by: java.lang.ArrayIndexOutOfBoundsException	1	Index 10 out of bounds for length 10, likely due to an incorrect array size or a flaw in the sorting algorithm logic.	Review the `Sorter.quickSort` method carefully, paying attention to array bounds and input sizes. Verify that the input array is of the expected length before attempting to sort it. Debug array access to identify the source of the out-of-bounds exception. **
java.lang.RuntimeException	1	Failed to process request – the root cause is not immediately clear from this trace alone, but likely involves an issue in the `OrderService.processOrder` method or a controller (e.g., `OrderController.submitOrder`).	Examine the code within `OrderService.processOrder` and the associated controller for potential exceptions or data validation failures. Add logging to capture relevant input parameters and internal state to aid in debugging. Review the overall order processing workflow for any potential inconsistencies or errors. **
java.lang.ArrayIndexOutOfBoundsException	1	Index 5 out of bounds for length 5 – indicates an issue with array access, probably within the `Sorter.quickSort` method or related code.	Double-check that the length of input array is correct before attempting to sort. Verify the logic used for determining the partition point in `quickSort` is accurate and doesn't result in an invalid index. **
java.lang.NullPointerException	1	`str` is null, preventing the call to String.length(). Likely a data issue where a string value is unexpectedly null, often related to user input or external data sources.	Implement robust null checks before attempting to perform operations that require non-null strings. Provide default values or handle null string inputs gracefully. Add validation to ensure that expected data is not missing and consider logging invalid data for further investigation.